widely distributed in North America, occurring from California to Alaska on the Pacific Coast. Studies on the distribution of ticks in western Oregon have shown that *I angustus* is the most common tick found in certain coastal regions of this area.[27] Although generally considered to be a nest-specific parasite of rodents and insectivores, *I angustus* is known to bite humans.[17] The wide distribution and common occurrence of this tick species may have important implications for potential Lyme disease transmission outside known endemic areas and an increased incidence within established areas. Furthermore, as our rural population expands and participation in outdoor recreational activities increases, placing humans into areas formerly the domain of wild animals and their parasites, exposure to ticks and tick-borne diseases will increase.

The transmission of Lyme disease spirochetes to humans may be minimized by the prompt and proper removal of a tick vector. The lengthy feeding period (several days) is an important factor contributing to the risk of infection.[28] In addition to timely removal, it is important that the mouthparts of an embedded tick be removed completely to prevent the occurrence of secondary bacterial infections. The procedure recommended for removing an attached tick is to grasp it as close to the skin surface as possible with forceps or protected fingers and gently pull backward with steady, even pressure.[29]

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Adrenal Apoplexy Revisited

KENNETH A. WOEBER, MD
San Francisco

DESPITE ITS INFREQUENCY, acute adrenocortical insufficiency is an important disorder to recognize because it is uniformly fatal unless promptly treated. Acute adrenocortical insufficiency is brought about by an intense stress occurring either in a patient with preexisting chronic adrenocortical insufficiency or in a patient with presumably normal adrenal glands that undergo massive hemorrhagic necrosis. This latter condition has been termed adrenal apoplexy. I describe the cases of two patients with adrenal apoplexy and discuss the cardinal aspects of its pathogenesis, clinical diagnosis, and management.

Report of Cases

Case 1

The patient, a 74-year-old woman, was admitted to the medical service because of fever, cough, shortness of breath, and chest pain for a week that had not responded to erythromycin therapy. She had had a resection of an adenocarcinoma of the sigmoid colon four years earlier without evidence of recurrence, a cerebrovascular accident six years earlier, and cutaneous lupus in the remote past at which time she was given glucocorticoids for three months. She was taking isosorbide dinitrate for mild angina and hydrochlorothiazide and triamterene for hypertension. On physical examination she was in moderate respiratory distress with a heart rate of 100 beats per minute, a blood pressure of 145/80 mm of mercury, a temperature of 38.3°C (101°F), a respiratory rate of 26 per minute, and signs of left lower lobe consolidation; the rest of the examination was unremarkable. A chest roentgenogram showed bilateral patchy infiltrates. Examination of a sputum specimen showed leukocytes and a mixture of bacteria that on culture yielded *Hemophilus influenzae* and oral flora. The blood leukocyte count was 8.1×10^9 per liter with 0.66 polymorphonuclear leukocytes and 0.17 band forms.

Treatment with intravenous fluids and cefamandole nafate was begun. On the fourth hospital day, the patient began

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From the Department of Medicine, Mount Zion Hospital and Medical Center and the University of California, San Francisco, School of Medicine.

Reprint requests to Kenneth A. Woeber, MD, Mount Zion Hospital and Medical Center, PO Box 7921, San Francisco, CA 94120-7921.

to have vague abdominal discomfort accompanied by nausea and occasional vomiting. These persisted, and despite clearing of the pulmonary infiltrates she continued to have daily temperature spikes to 39°C (102°F). She became progressively weaker and more lethargic. On the ninth hospital day, blood cultures showed no growth and the blood leukocyte count was 5.9×10^9 per liter with a normal differential. On a serum electrolyte panel, the sodium level was 122 mmol per liter and the potassium was 4.9 mmol per liter. Over the ensuing five days, the blood pressure ranged around 100/60 mm of mercury and the lethargy, nausea, and daily temperature spikes persisted. On the 14th hospital day, she had a cardiac arrest from which she was initially resuscitated, but despite maximal support with pressors, hypotension persisted and she died.

A necropsy examination disclosed organizing bilateral pneumonitis, extensive infiltration of para-aortic lymph nodes by adenocarcinoma similar in histopathologic appearance to the previously resected colonic carcinoma, and extensive hemorrhage within both adrenal glands (Figure 1). On histopathologic examination of both adrenal glands, there was extensive hemorrhagic necrosis with thrombi in the adrenal vein and its tributaries but no evidence of metastases.

Case 2

A 69-year-old man was admitted to the medical service because for ten days he had had weakness, intermittent nausea and vomiting, mental confusion, and fever. Two weeks earlier he had undergone right- and left-sided catheterization of the heart that apparently was uncomplicated. For three years he had had congestive heart failure accompanied by intermittent atrial fibrillation. His blood pressure, reportedly, was always in the range of 130/80 mm of mercury. He had not taken glucocorticoids. He was taking verapamil hydrochloride, 80 mg three times a day.



Figure 1.—The transected adrenal gland of patient 1 shows a central hematoma (courtesy of Harvey Z. Klein, MD, Department of Pathology and Laboratory Medicine, Mount Zion Hospital and Medical Center).

On physical examination he was lethargic with a blood pressure of 110/70 mm of mercury while supine that fell to 100/60 mm of mercury on sitting, a temperature of 38°C (100.4°F), and cardiomegaly with a systolic ejection murmur along the left sternal border; there were no signs of congestive heart failure. Specimens of blood, urine, and cerebrospinal fluid were cultured, and treatment with penicillin, nafcillin sodium, and tobramycin was begun for a presumed infection. There was no growth on the cultures after three days, and the antibiotic therapy was withdrawn. The fever and lethargy persisted, however, and because of a serum sodium level of 123 mmol per liter and a potassium level of 4.6 mmol per liter, a cosyntropin stimulation test was done. The serum cortisol level was 110 nmol per liter (4 μg per dl) before and 140 nmol per liter (5 μg per dl) one hour after cosyntropin was administered. Accordingly, an intravenous infusion of hydrocortisone was begun and within 12 hours the fever and lethargy had disappeared and the blood pressure had risen to 130/80 mm of mercury. A computed tomographic scan of the abdomen showed massive enlargement of both adrenal glands consistent with hemorrhage. Treatment with oral hydrocortisone and fludrocortisone acetate was instituted, and at the time of discharge his blood pressure and serum electrolyte values were normal.

Discussion

In several large necropsy series, the incidence of adrenal apoplexy—that is, massive adrenal hemorrhagic necrosis—ranged between 0.16% and 0.69%.[1-5] These figures, however, underestimate the true incidence of adrenal apoplexy because in the series cited, the denominator was not restricted to patients dying of shock or in whom death was sudden and unexpected. Thus, the diagnosis is still often left to a pathologist, as exemplified by the case of patient 1. On the other hand, in the case of patient 2, even though the antecedent stressful insult appeared minor, prompt diagnosis and therapeutic intervention led to full recovery.

The clinical setting in which adrenal apoplexy usually occurs is that of stress accompanying a major insult, such as severe infection, a perforated viscus, trauma, burns, eclampsia, or a cardiovascular catastrophe. Anticoagulant therapy is an occasional contributing factor.[2] It has also been reported to occur as a complication of treatment with adrenocorticotropin (ACTH).[6] At necropsy, the adrenal glands present a characteristic appearance consisting of a central hematoma enveloped by a rim of residual cortex and of thrombosis of the adrenal vein and its tributaries.[1] These features were present in the case of patient 1 (Figure 1).

The pathogenetic mechanism underlying adrenal apoplexy is still enigmatic. It may be sought, however, in the unique vascular anatomy and special functional characteristics of the adrenal gland. Several authors[4,7] have drawn attention to the fact that the adrenal gland is endowed with an exceptionally rich arterial blood supply that increases remarkably during stress. By contrast, venous drainage is limited to thin-walled venous sinuses that drain into a single vein. A fairly consistent finding in the adrenal gland that is the seat of hemorrhage is the presence of thrombi in the adrenal vein and its tributaries.[1,3,4,7,8] It has been suggested

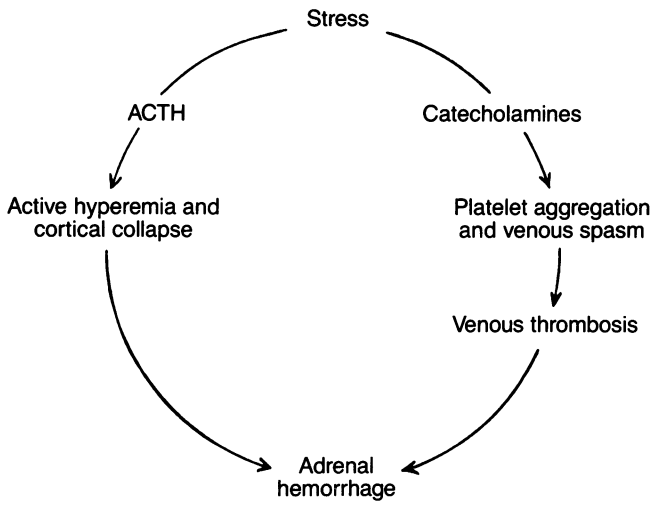


Figure 2.—The schema shows the putative pathogenetic interplay that leads to adrenal hemorrhage.

that the high concentrations of cortisol and catecholamines in the adrenal venous effluent during stress induce venous spasm and platelet aggregation, thereby leading to thrombus formation.[8] Moreover, ACTH stimulation of the adrenal gland during stress leads to hyperemia and to lipid depletion that favors cortical collapse. In this regard, it is noteworthy that adrenal apoplexy may follow treatment with exogenous ACTH.[6] Thus, the interplay of all the foregoing factors might ultimately result in adrenal hemorrhage (Figure 2). Indeed, the characteristic gross pathologic finding of a massive central hematoma is entirely consistent with this pathogenetic interplay.

The clinical presentation is nonspecific.[4] Accordingly, acute adrenocortical insufficiency resulting from adrenal apoplexy should be considered in any patient in whom hypotension or vascular collapse supervenes unexpectedly or whose condition does not respond to the usual supportive measures. Vomiting and abdominal discomfort may occur and in some instances be severe enough to mimic a surgical acute abdomen. Occasionally, adrenal apoplexy may be accompanied by flank or back pain. The presence of hyponatremia in association with azotemia should enhance one's index of suspicion although hyponatremia is not a consistent feature, and hyperkalemia is often not present.[9] As exemplified by the case of patient 2, a computed tomographic scan of the abdomen would suggest the presence of adrenal apoplexy if bilateral adrenal enlargement were found.[10]

The diagnosis of acute adrenocortical insufficiency in an acutely ill patient is readily established by measuring the serum cortisol level. In this circumstance, a value that is not clearly increased strongly suggests a diminished adrenocortical reserve, and a diminished or absent response to cosyntropin would provide absolute confirmation. Once a blood specimen has been collected, treatment with intravenous hydrocortisone in a dosage of 200 mg to 300 mg during the first 24 hours should be instituted. Such an approach would salvage patients from a life-threatening disorder that is eminently treatable if recognized early.

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Postpartum Immunologic-Mediated Pulmonary Edema Associated With Transfusion of Blood Containing an Anti-B-Lymphocyte Antibody

RICHARD S. NEWMAN, MD
JAMES H. WILLIAMS, MD
LESLIE J. MOBERG, MD
MARIE L. COOK, MT, SBB
Irvine, California

POSTTRANSFUSION IMMUNOLOGIC-MEDIATED PULMONARY EDEMA is a condition of acute respiratory failure that is most frequently caused by an antibody in donor blood that has a specificity toward an antigen found on patient granulocytes, lymphocytes, or both.[1] We describe here a postpartum case in which low-dose steroid treatment led to a dramatic improvement in oxygenation.

Report of a Case

The patient, a 24-year-old woman, gravida 3, para 2, was admitted at 37½ weeks' gestation with placenta previa. Her medical history was notable for heroin abuse, and she was now on methadone maintenance. Admission laboratory values included a leukocyte count of 11.4×10^9 per liter, a hematocrit of 0.28, and a platelet count of 204×10^9 per liter. The admission differential included 0.74 segmented polymorphonuclear leukocytes, 0.01 band forms, 0.01 metamyelocytes, 0.10 eosinophils, 0.11 lymphocytes, and 0.03 monocytes. A test for the hepatitis B surface antigen was negative.

The patient underwent a cesarean section with an estimated blood loss of 1,500 ml, and she was given a transfusion of two units of erythrocytes. While in the recovery room, she had tachypnea, tachycardia, and hypotension with a blood pressure of 85/44 mm of mercury. On examination of the lungs, there were no rales. Blood gas determinations done with the patient breathing 6 liters of oxygen by mask indicated a Po_2 of 5.86 kPa (44 mm of mercury), a Pco_2 of 4.26 kPa (32 mm of mercury), a bicarbonate of 22 mmol per

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From the Departments of Pathology (Dr Newman and Ms Cook), Medicine (Dr Williams), Obstetrics and Gynecology (Dr Moberg), University of California, Irvine, California College of Medicine, Irvine, and the University of California, Irvine, Medical Center, Orange, California.

Reprint requests to Richard S. Newman, MD, Department of Pathology, Rt 40, UCI Medical Center, Orange, CA 92668.